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Q10



Q3

$$2x^2 - 5x + 3 = 0$$

$$x = [5 \pm \sqrt{25 - 24}] / 4 = (5 \pm 1) / 4$$

$$\text{so } x = 1.5 \text{ or } x = 1$$

Q1

Inertia is the tendency of a body to resist any change in its state of rest or of uniform motion.



Q7

Ice floats because it is lighter than water.

Ice floats on water because ice is less dense than liquid water - the hexagonal crystal structure formed on freezing takes up more space than the liquid.

Q11

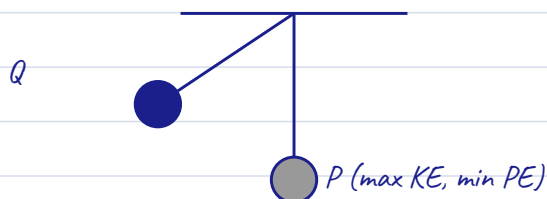
Ice is less dense so it floats on top of water.

Q1

Mitochondria is the powerhouse of the cell.

Q4

P = lowest point (max speed, max KE, min PE). Q and R are the extreme points where the bob momentarily stops (max PE, KE = 0). As the bob moves from R to P, PE converts into KE.



Q8(a)

Specific heat capacity is the heat needed to raise the temperature of 1 kg of a substance by 1 degree Celsius.

Q8(b)

$$Q = mc(\Delta T) = 2 \times 4200 \times 60 = 504000 \text{ J}$$

Q11(b)

In molten NaCl the ions are free to move and carry charge but in solid NaCl the ions are fixed in the lattice and cannot move so it cannot conduct electricity.

Q20

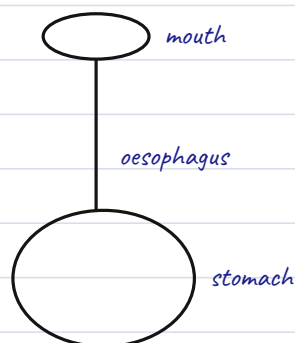
The mitochondria produces ATP through cellular respiration.

Q9

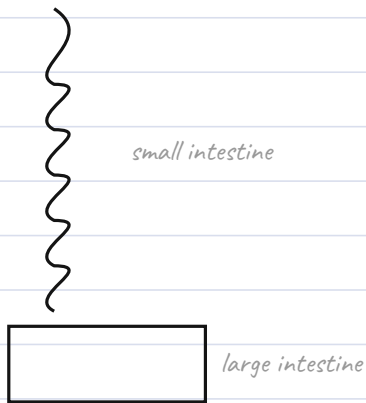
1. Plant cells have a cell wall, animal cells do not.
2. Plant cells have chloroplasts and a large vacuole, animal cells usually do not.

Q12

(diagram - human digestive system)



Q12 (contd.)



Q2

The SI unit of force is the newton (N). One newton is the force required to give a mass of 1 kg an acceleration of 1 m/s^2 .

Q14

$$x^2 - 7x + 10 = 0$$

$$x = [7 \pm \sqrt{49 - 40}] / 2 = (7 \pm 3) / 2$$

$$\text{so } x = 5 \text{ or } x = 2$$

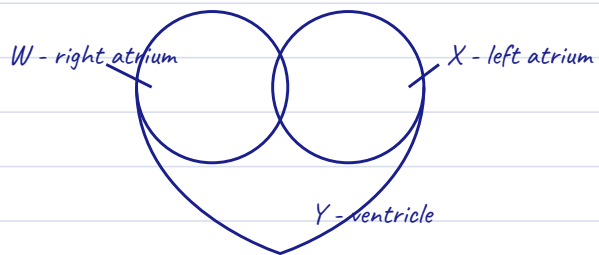
Q16

$$\text{Volume} = \pi r^2 h = \frac{22}{7} \times 7 \times 7 \times 20 = 3080 \text{ cm}^3$$

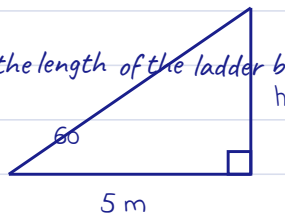
$$\text{TSA} = 2 \pi r(r+h) = 2 \times \frac{22}{7} \times 7 \times 27 = 1188 \text{ cm}^2$$

Q13

(diagram - heart, chambers marked below)



Q15

Let the length of the ladder be L .

$$\cos 60 = 5 / L \Rightarrow L = 5 / \cos 60 = 5 / 0.5 = 10 \text{ m}$$

Q18

$$96 = 2^5 \times 3$$

$$404 = 2^2 \times 101$$

$$\text{HCF} = 2^2 = 4$$

$$\text{LCM} = 2^5 \times 3 \times 101 = 9696$$

ROUGH WORK*(This page will not be evaluated)*

147x... ^{no} wait

2x5=10 3x5=15 4x5=20



22/7 * 7 = 22 ... check

Sqrt2 = 1.41421...